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A Low-power High-impedance Instrumentation Amplifier with Capacitor-calibration for Bio-sensors

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***Abstract*—The analog front-end (AFE) in biomedical sensors requires high input impedance to suppress interference during bio-potential acquisition. This paper presents a low-power chopping-stabilized AFE with ultra-high input impe-dance. A self-adjusting variable-capacitor in the positive feed-back loop (PFL) dynamically compensates parasitic capacitan-ce, while an oscillation detection loop guides the system from instability to stability, maximizing impedance without over-compensation. A frequency ripple suppression method further mitigates input frequency fluctuations, providing accurate control signals for the variable capacitor. This capacitor decouples control and input signals, ensuring high linearity and wide tuning range. The technique achieves femtofarad-level resolution, overcoming limitations of traditional capaci-tive arrays in PFL. Fabricated in 0.18 µm CMOS, the AFE attains 46.3 GΩ input impedance at 50 Hz—83× higher than conventional designs—occupying 0.71 mm² with 40 dB mid-band gain and only 68 fF input capacitance. It also achieves low noise (0.93 µVrms) and low power consumption (3.48 µW), extending acquisition system operation. Using dry electrodes, the prototype successfully acquired high-quality EEG and ECG signals, validating its applicability to biomedical devices.**

Keywords—analog front-end, bio-medical sensors, high input impedance, variable-capacitor calibration

I. INTRODUCTION

Bioelectric signals, including electrocardiograms (ECG), electroencephalograms (EEG), and electromyograms (EMG), provide real-time insight into the physiological state of the human body [1]. These signals are essential for personalized telemedicine, intelligent health management, and unmanned control systems [2], [3]. They are acquired through contact electrodes, transmitted via electrode leads to the analog front-end (AFE) for amplification [4], and then forwarded to the cloud or a smart terminal for real-time visualization. As the core component of this process, the AFE plays a critical role in ensuring the integrity of the entire readout system.

Numerous techniques have been proposed to improve AFE signal quality [5]. These include carefully designed printed circuit boards (PCBs) and negative feedback to suppress common-mode interference [6], differential-diffe-rence amplifiers and OTA-stacking to reduce noise [7], as well as common-mode duplication and differential regulation to enhance CMRR [8]. However, most rely on conventional components, resulting in high system complexity and power consumption [9]. Moreover, the AFE's low input impedance, caused by capacitive coupling between electrodes and skin, limits anti-interference capability [10]. Since AFEs are widely integrated into wearable devices [11], [12] (e.g., watches and glasses), diverse interface conditions can introduce significant parasitic capacitance at the input nodes.

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Within the target frequency band, the AFE must maintain high input impedance to minimize signal attenuation and adapt to contact variations. Although various methods have been explored to boost input impedance [13], achieving input impedance beyond 10 GΩ at 50/60 Hz remains challenging.

To address these issues, this paper proposes an ultra-high input impedance AFE for bioelectric signal acquisition. The design employs a positive feedback loop (PFL) with vari-able-capacitor calibration (VCC) that automatically adjusts feedback strength based on oscillation amplitude. An oscilla-tion detection block guides the AFE from instability to steady operation, maximizing input impedance without com-promising stability. Meanwhile, a continuously tunable capa-citor with ultra-high resolution effectively cancels parasitic capacitance at the input front-end. The remainder of this paper is organized as follows. Section II describes in detail the proposed calibration mechanism for high input im-pedance and the circuit implementation of AFE. Section III shows the proposed AFE acquisition prototype and the measurement results. Section IV summarizes this work.

II. IMPLEMENTATION OF THE PROPOSED AFE

A. Overall Structure

The proposed high impedance AFE, as shown in Fig. 1, comprises the following components: 1) a two-stage Miller-compensated OTA with chopper modulation, labeled as G_{m1} and G_{m2}; 2) a DC servo loop (DSL) and a ripple reduction loop (RRL); 3) a PFL with a variable capacitor C_{IBL}; 4) an amplitude detection block (amplitude detection, subtractor, and integrator), which is used to generate the control voltage V_C. The closed-loop gain of the AFE is determined by the ratio of the input capacitance C_{IN} to the feedback capacitance C_{FB} and is fixed at C_{IN} / C_{FB} = 100.

In the proposed AFE, the chopper structure allows the flicker noise of the two-stage operational amplifiers to be shifted from the baseband to high frequencies. The use of a complementary input-based G_{m1} effectively doubles the

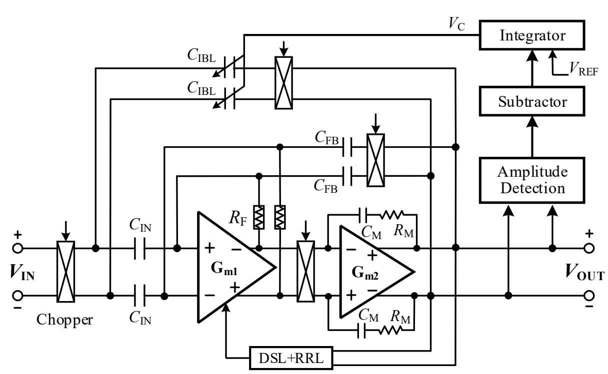

Fig. 1. Proposed overall architecture for high input impedance AFEs.

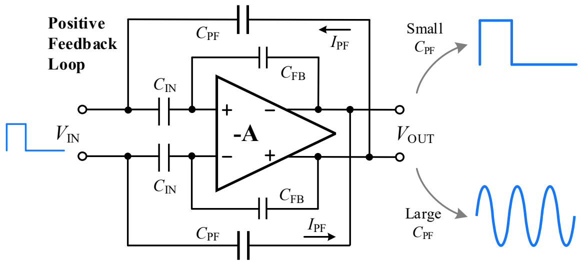

Fig. 2. AFE transient response of with a positive feedback loop.

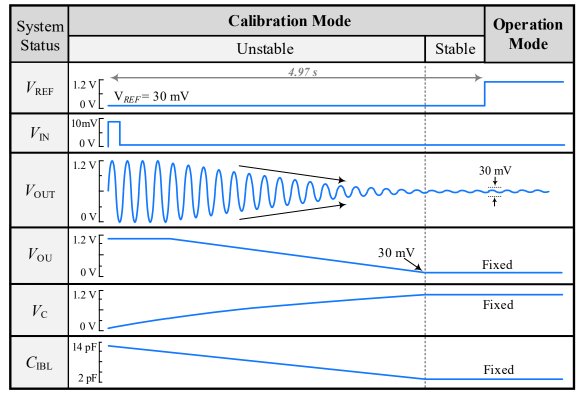

Fig. 3. Timing diagram of AFE in calibration and operation modes.

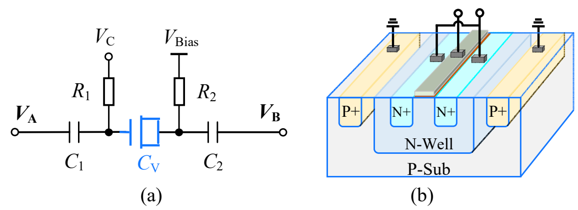

Fig. 4. (a) Circuit structure of C_{IBL}, (b) 3-D structure of varactor C_V.

transconductance without increasing the bias current. A structure with complementary inputs can reduce the IRN by a factor of $\sqrt{2}$ compared to no complementary inputs [14]. The large resistor R_F, connecting the input and output of G_{m1}, serves as a DC feedback loop and establishes a stable DC bias for the input differential pair. Notably, G_{m2} achieves high linearity with smaller input swing amplitude. As the noise of G_{m2} is much lower than that of G_{m1}, G_{m2} can be biased at a relatively lower current for better power efficiency. The implementation of Miller compensation ensures the stability of the closed loop. Additionally, common-mode feedback employs a resistive sensing scheme to further enhance linearity.

B. Principle of the Proposed Ultra-high Input Impedance

As shown in Fig. 2, the proposed AFE achieves marginal stability by dynamically adjusting the feedback capacitor C_{IBL} according to the output oscillation condition, thereby compensating for the input parasitic capacitance and maximizing input impedance. Amplifiers based on the PFL technique determine stability by observing the output response to an input pulse [15]. When C_{PF} is comparable to the feedback capacitance C_{FB}, the AFE operates stably as an amplifier; when C_{PF} exceeds C_{FB}, excessive positive feedback causes ringing and oscillation. Thus, C_{PF} is adjusted based on PFL stability criteria to achieve optimal parasitic compensation.

To realize this adaptive mechanism, a VCC scheme is proposed to be integrated into the AFE. The variable-capacitor C_{IBL} provides controllable positive feedback gain, tuned by a control voltage V_C generated by an amplitude detection block comprising detectors, a subtractor, and an integrator. The block detects output peaks and valleys, calculates the amplitude, compares it with a reference voltage V_{REF}, and integrates their difference to produce V_C. V_C then adjusts the feedback strength until the AFE transitions from oscillation to stability.

The AFE operates in two modes: calibration and normal operation. V_C varies only during calibration, as illustrated in Fig. 3. When output oscillations reach rail-to-rail levels, V_C gradually increases, reducing positive feedback and forcing C_{IBL} toward matching the input parasitic capacitance, thereby enhancing input impedance T_{ZIN}. Once the oscillation amplitude drops below V_{REF} = 30 mV, V_C stabilizes, fixing C_{IBL} and concluding the calibration. Because C_{IBL} is continuously rather than discretely adjustable, the proposed design achieves convergence within 5 s—significantly shorter than the >30 s calibration period reported in [16].

C. Structure of Variable Capacitors

A chopper-modulated variable-capacitor loop is incorporated into the positive feedback path of the AFE, converting the output voltage into a feedback current that compensates for the currents drawn by the chopper, input capacitor, and parasitic capacitance, thereby significantly enhancing the input impedance. The proposed PFL topology with continuously tunable variable capacitors offers three key advantages: (1) voltage-controlled capacitors provide ultra-high resolution beyond traditional capacitor array limits; (2) they introduce no additional noise or gain mismatch, ensuring compatibility with various closed-loop gains; and (3) their fixed layout minimizes parasitic variations caused by capacitor switching in conventional array structures.

The continuously adjustable C_{IBL} is depicted in Fig. 4(a), consisting of an accumulation-type MOS varactor diode C_V, MIM capacitors C_1, C_2, and resistors. V_A and V_B are the signal nodes, and V_C is the control node. Decoupling the signal nodes from the control node overcomes the limitation of traditional 2-T variable-capacitors whose capacitance values change with the output signal. The 3-D structure of C_V is shown in Fig. 4(b). The capacitance looking from the gate is simply the oxide capacitance. Therefore, there is a first-order relationship between the capacitance values C_V and V_{GS} in the near flat-band voltage:

$$C_V = \beta \cdot (V_C - V_{Bias}) \tag{1}$$

where β is a first-order coefficient. Under the control of gate voltage V_C, the oxide capacitance of C_V varies with the concentration of electrons and impurities. The capacitance C_V is inversely related to V_C and directly related to the bias voltage V_{Bias}. The simulation results clearly demonstrate that the implemented C_{IBL} exhibits excellent controllability and good linearity, as expected. By changing V_{Bias}, the capacitance value can be tuned and shift the C-V curve left or right. The V_{Bias} is selected to be 0.6 V depending on the operating voltage requirements of the AFE. A continuously adjustable capacitance value exceeding 14 pF is achieved under a wide voltage range.

Fig. 5 displays the layout of the variable-capacitor. The C_V is positioned directly below the MIM capacitor, and the critical wiring is shielded with metal from adjacent layers to minimize the parasitic capacitance of the connections.

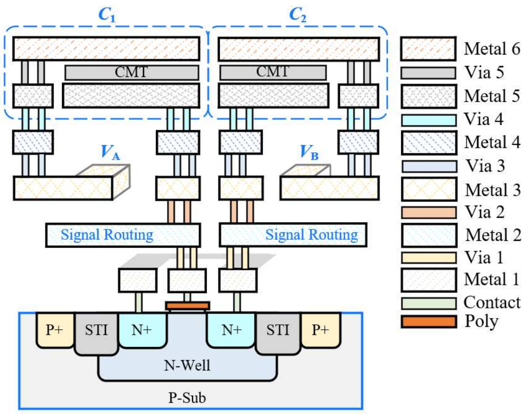

Fig. 5. Layout structure of the implemented variable-capacitor C_{IBL}.

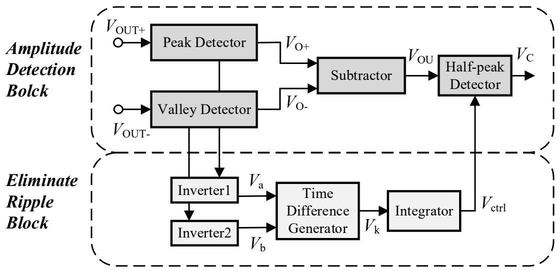

Fig. 6. Amplitude detection block with ripple elimination technique.

D. Amplitude Detection Block With Ripple Elimination

The V_C, which can precisely reflect the oscillation amplitude information of an AFE output, is crucial for tuning the C_{IBL}. Considering that the frequency of bioelectrical signals varies continuously in practical scenarios, this results in the inevitable production of ripple in the detected amplitude. This occurs because the charge on the capacitor C is constantly changing, and the current source I_B used for dis-charging cannot adjust at a corresponding rate. Additionally, the output voltage periodically decreases due to charge leakage. The ripple caused by the mismatched charging and discharging speeds affects the accuracy of the V_C.

A ripple elimination scheme is integrated into the amplitude detection block, as shown in Fig. 6. It consists of an inverter, a time-difference generator [17], and a half-peak detector. The inverter monitors signal nodes in the peak and valley detectors correlated with the input frequency. The resulting square wave is processed by the time-difference generator to produce a pulse waveform whose duty cycle controls the charging and discharging rates of the half-peak detector's current source. Thus, the detector automatically adjusts its response to the input frequency, remaining unaffected by signal amplitude.

Simulation results of the detection block under different frequencies and amplitudes are shown in Fig. 7. The differential signals V_{IN+} and V_{IN-} vary as 0.32 V@10 Hz (0.025–0.225 s), 0.65 V@20 Hz (0.225–0.375 s), and 1 V@40 Hz (0.375–0.525 s), followed by a gradual amplitude decrease at 80 Hz. Without the ripple elimination technique, the detected amplitudes exhibit pronounced low-frequency ripples caused by charge–discharge mismatch, resulting in unstable feedback gain and degraded capacitor linearity. In contrast, the proposed technique significantly suppresses ripples, enabling accurate amplitude detection across varying frequencies. The comparison of output waveforms confirms that the ripple elimination block effectively enhances detection accuracy and ensures precise extraction of AFE oscillation states.

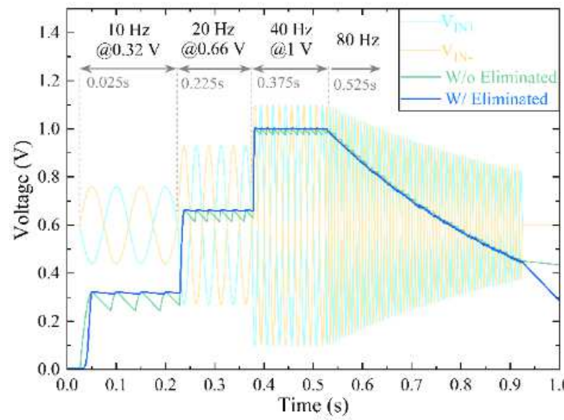

Fig. 7. Comparison of amplitude detection results, where V_{IN+} and V_{IN-} represents the differential output. The ripple with frequency elimination technique is significantly reduced.

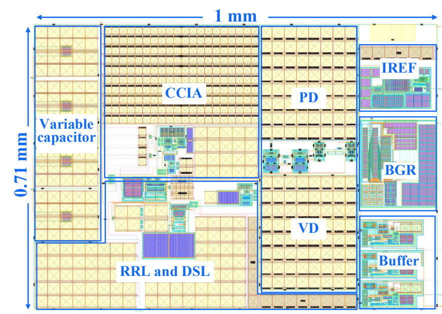

Fig. 8. Layout of the high-input-impedance AFE chip.

III. Measurement Results

The proposed high impedance AFE is fabricated using a standard 0.18 μm CMOS process, and the layout is shown in Fig. 8. The effective chip area is 0.71 mm^2 with the optimal matching using dummy technique. The AFE consumes 2.9 μA at 1.2 V supply, where the two-stage OTA dominates the power consumption due to its low noise requirement.

A. Input Impedance

Before impedance measurement, the AFE performs self-calibration to reach the optimal impedance state. Once the control voltage V_C stabilizes, the circuit enters operational mode for bio-signal acquisition. All measurements are conducted in a metal shielding box to minimize electromagnetic interference. As shown in Fig. 9(a), the proposed self-calibrated VCC technique dramatically improves impedance performance compared with the conventional PFL. The measured input impedance increases from 554 MΩ to 46.3 GΩ at 50 Hz—an 83× enhancement—corresponding to an equivalent capacitance of only 68 fF. Within the frequency range of interest, the AFE maintains impedance above 100 GΩ at 1 Hz, significantly reducing signal attenuation caused by source voltage division. To further verify adaptability under different electrode contact conditions, Fig. 9(b) compares impedance versus parasitic capacitance. With increasing C_{P-EXT} from 1 pF to 9 pF, the traditional PFL suffers severe degradation (478 MΩ to 55 MΩ), while the proposed

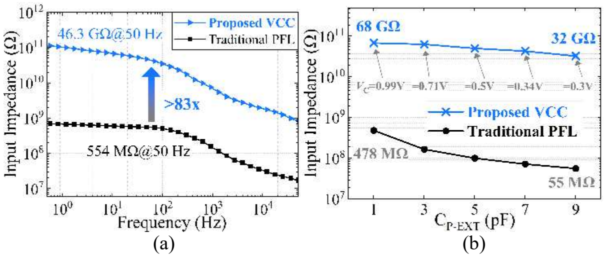

Fig. 9. Measurement results of (a) input impedance versus frequency, and (b) input impedance at 20 Hz with various $C_{P\text{-}EXT}$s.

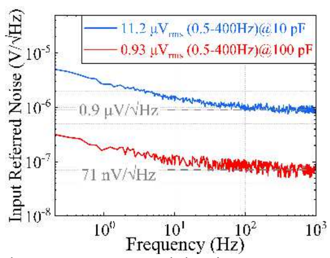

Fig. 10. Measured IRN power spectral density.

VCC technique maintains >32 GΩ by dynamically adjusting C_{IBL} and V_C. This adaptive compensation ensures high and stable impedance across varying conditions, making the proposed AFE highly suitable for precise signal acquisition.

B. Characteristics of the Proposed AFE

The AFE measured a midband gain of 39.9 dB, with 0.35 Hz high-pass and 5.4 kHz low-pass cutoff frequencies. As shown in Fig. 10, input-referred noise was measured using external capacitors of 10 pF and 100 pF to emulate varying electrode contact conditions. Within 0.5–400 Hz, the integrated noise was 11.2 µVrms and 0.93 µVrms, respectively, while above 100 Hz, the thermal noise densities were 0.9 µV/√Hz and 71 nV/√Hz. These results confirm excellent noise performance under high input impedance operation. The proposed AFE achieves a CMRR of 81 dB under a 0.6 Vpp common-mode input, benefiting from its ultra-high input impedance and chopper stabilization. The measured PSRR under a 0.1 Vpp supply ripple reaches 77 dB, indicating strong immunity to supply noise.

C. In Vivo Measurement

The proposed AFE was evaluated through two in vivo tests: ECG acquisition using dry electrodes and EEG acquisition using ear-clip electrodes. As shown in Fig. 11, ECG signals were measured with 1 cm² dry electrodes coupled to the skin through clothing. Signals were transmitted via shielded coaxial cables to an oscilloscope (AC coupling, 20 kSa/s, 500 Hz bandwidth) to preserve waveform integrity. Fig. 12 displays the recorded ECG, showing clear QRS complexes without saturation or noticeable power-line or

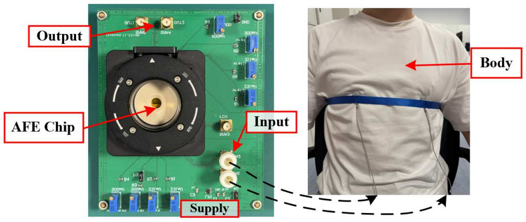

Fig. 11. AFE prototype uses non-contact electrodes for ECG acquisition.

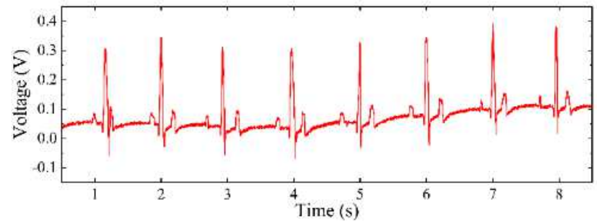

Fig. 12. Raw data results of the acquired ECG signal.

myoelectric interference. The slight baseline drift results from loose electrode contact through clothing, which can be mitigated by adjusting electrode placement while maintaining user comfort. The spectral results using comfortable and repeatable ear-clip electrodes reveal a distinct 11 Hz alpha rhythm during eyes-closed conditions, confirming high feature resolution and low noise floor.

D. Performance Summary and Comparison

Table I summarizes the performance of the proposed AFE compared with other IAs for biopotential acquisition. Previous studies [5], [19] improved input impedance using active shielding but at the cost of higher power and degraded noise efficiency. Works such as [16], [20] achieved lower noise at wider bandwidths but suffered from large input equivalent capacitance due to traditional PFL. The proposed AFE has an input equivalent capacitance of 68.7 fF, compa-rable to the state-of-the-art design in [13], with similar noise efficiency and lower overall power consumption.

IV. Conclusion

This paper presents a high-input-impedance AFE for biosensors. Equipped with an oscillation detection circuit and a ripple elimination block, it precisely compensates for input parasitic capacitance and provides accurate control for the variable capacitor. The adjustable capacitor rapidly stabilizes during calibration without affecting circuit stability. Implemented in a standard 0.18 µm CMOS process, the AFE consumes 3.48 µW with an area of 0.71 mm², achieving an input impedance of 46.3 GΩ at 50 Hz and an IRN of 0.93 µVrms over 0.5–400 Hz. In-vivo ECG and EEG tests confirm its suitability for low-power, reliable physiological signal acquisition for continuous health monitoring.

TABLE I
Performance Comparison With The State-of-the-art AFEs

Parameter	JSSC '17 [18]	TBCAS '19 [5]	ISSCC '20 [19]	JSSC '22 [16]	JSSC '24 [13]	VLSI '24 [20]	**This Work**
Power (µW)	2.8	90	3.35	3.83	10.46	13.86	**3.48**
Area (mm²)	0.069/ch	1.2	0.22	0.75	0.19	0.085	**0.71**
IRN (μV_{rms})@ Hz	1.8(1–200)	3.7(0.5–100)	3.15(0.5–5k)	0.36(0.5–300)	0.72(0.5–100)	0.59(1–100)	**0.93(0.5–400)**
Impedance(GΩ) @Hz	1@50	4@50	50@50	2@50	136@60	0.064@60	**46.3@50**
Self-calibration	No	No	No	Yes	Yes	No	**Yes**
$C_{in,eq}$ (fF)	3184	796	63	1592	19.5	41472	**68.7**
Boosting Techniques	Pre-charge	Active Shield & Bootstrap	Active Shield & Trimming PFL	Dual PFL	Active Shield & C_{PF} Down Scaling	Small Input Capacitance	**Variable-capacitor Calibration**

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